

Constraining Dark Energy Equation of State with High-Redshift LSST Stacked Image Mocks

Input Question

When will the universe die? Will it continue to expand?

Domain

Astronomy

Validation Status

needs_data

Problem Statement

The ultimate fate of the universe and the continuity of its expansion depend on the properties of dark energy. Current understanding, as noted in the source material, is insufficient to definitively predict the timeline or final state, requiring better constraints on cosmological parameters derived from high-redshift observations.

Rationale

Evidence EV-Q050-bc54ad36646f99e2f84349ca indicates that the LSST survey is designed to study dark energy and the accelerating expansion of the Universe by detecting galaxies to redshifts well beyond unity. Improving constraints on the dark energy equation of state parameter (w) using these high-redshift data is a critical step toward reducing uncertainties in cosmological models that predict the universe's expansion history.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: LSST stacked images detecting galaxies at redshifts well beyond unity will significantly reduce statistical uncertainties on the dark energy equation of state parameter w compared to current baselines, thereby providing necessary (but not sufficient) observational constraints for future theoretical determination of universal expansion fate.
- **Mechanism**: EV-Q050-bc54ad36646f99e2f84349ca establishes that LSST is designed to study dark energy and accelerating expansion via stacked images reaching $z > 1$. Since dark energy properties govern expansion dynamics, improving empirical constraints on w using this specific observational capability is a prerequisite for addressing the knowledge gap regarding expansion continuity. This hypothesis strictly limits the claim to 'improving constraints' as supported by the evidence's design intent, explicitly treating the mapping from w to ultimate fate as a pending theoretical validation step rather than an established mechanism.
- **Falsifiable Prediction**: If analysis of LSST simulated stacked image mocks consistent with EV-Q050-bc54ad36646f99e2f84349ca specifications yields a marginalized constraint on w that is statistically

indistinguishable from or wider than current pre-LSST baselines (e.g., DES Y3 or Planck), then the hypothesis that LSST stacked images provide significant improvement is falsified.

- **Required Observations**: Galaxy number counts and clustering statistics derived from LSST simulated stacked image mocks at $z > 1$; Posterior probability distribution of w derived from joint analysis of LSST mock data ; Comparative Figure of Merit (FoM) or σ_w values against established pre-LSST baselines
- **Risk of Being Wrong**: Systematic errors in photometric redshift estimation for faint sources in stacked images may dominate over statistical gains, negating the expected improvement; or the assumed survey depth/duration in mocks may exceed actual operational performance described in EV-Q050-bc54ad36646f99e2f84349ca.

Hypothesis 2

- **Hypothesis**: The high-redshift reach ($z > 1$) of LSST stacked images is the critical factor enabling improved dark energy constraints, such that excluding this redshift range eliminates any statistical advantage over existing surveys.
- **Mechanism**: EV-Q050-bc54ad36646f99e2f84349ca specifically highlights the capability to detect galaxies 'to redshifts well beyond unity' as a feature of the stacked images. This hypothesis tests whether this specific evidentiary claim translates into tangible cosmological utility. If the high- z data does not break degeneracies or improve leverage on w , then the specific design feature mentioned in the evidence is not effective for this science goal.
- **Falsifiable Prediction**: If an ablation study comparing w constraints from full LSST mocks versus mocks truncated at $z < 1$ shows no statistically significant difference in uncertainty or FoM, then the hypothesis that high- z reach is critical is falsified.
- **Required Observations**: w constraints derived from full LSST simulated stacked image mocks ; w constraints derived from identical mocks with $z > 1$ data removed ; Statistical comparison (e.g., ratio of uncertainties) between the two samples
- **Risk of Being Wrong**: High- z galaxies may have higher systematic biases that are not properly modeled, leading to apparent but spurious improvements; or low- z data may already saturate the constraint making high- z redundant for constant w models.

Technical Details

This experiment evaluates the potential of LSST stacked images to constrain the dark energy equation of state parameter w relative to current observational baselines. Per EV-Q050-bc54ad36646f99e2f84349ca, LSST is designed to detect galaxies to redshifts well beyond unity ($z > 1$) via stacked images. The technical approach involves generating simulated galaxy catalogs consistent with these LSST design specifications (mock data), rather than using specific named simulations not present in the evidence. We employ a joint likelihood analysis of two-point correlation functions (galaxy clustering and weak lensing shear) in tomographic redshift bins. The parameter estimation uses standard cosmological notation: σ_w denotes the statistical uncertainty on w , and Ω_m denotes matter density. The Friedmann equations are used to map cosmological parameters to observable distances, but the mapping from w to the ultimate 'fate' of the universe is explicitly treated as a theoretical knowledge gap requiring future validation, not an established result of this experiment. Systematic errors in photometric redshifts for high- z sources are marginalized over using nuisance parameters.

Datasets

Source

```
```json
{
 "name": "LSST Simulated Stacked Image Mocks",
 "description": "Synthetic galaxy catalogs generated to match the LSST design specifications described in EV-Q050-bc54ad36646f99e2f84349ca, specifically ensuring coverage of galaxies at redshifts $z > 1$. These mocks serve as proxies for the planned decade-long survey data.",
 "access_note": "Generated internally or sourced from public mock challenges consistent with LSST design; not yet real observational data.",
 "evidence_ids": [
 "EV-Q050-bc54ad36646f99e2f84349ca"
],
 "is_public_candidate": true,
 "is_already_downloaded": false
}
```
```

Target

```
```json
{
 "name": "Constraints on Dark Energy Equation of State (w)",
 "description": "Posterior distribution of w derived from LSST mocks, compared against pre-LSST baselines. The primary metric is the reduction in σ_w (uncertainty) relative to current surveys.",
 "type": "derived_parameter"
}
```
```

Paper Abstract

Background: The ultimate fate of the universe depends on the nature of dark energy, which drives the accelerating expansion of the Universe. Evidence suggests that future surveys like LSST are designed to probe this acceleration by observing galaxies at high redshifts ($z > 1$). Methods: We propose a simulation-based study using mock data generated according to LSST design specifications (EV-Q050-bc54ad36646f99e2f84349ca). We perform a joint

likelihood analysis of galaxy clustering and weak lensing shear to constrain the dark energy equation of state parameter w . Validation Plan: We compare the resulting constraints (σ_w) against current baselines such as Planck and DES Year 3. An ablation study excludes $z > 1$ data to test the specific value of high-redshift observations. Results: pending (待执行验证实验).

Methods

1. **Mock Data Generation**: Create lightcone mocks reflecting LSST's capability to detect galaxies at $z > 1$ as per EV-Q050-bc54ad36646f99e2f84349ca. 2. **Statistical Analysis**: Compute angular power spectra for galaxy clustering and weak lensing shear in multiple redshift bins. 3. **Parameter Inference**: Use MCMC methods to sample the posterior distribution of cosmological parameters, including w . 4. **Comparative Assessment**: Compare σ_w from LSST mocks against constraints from existing surveys. 5. **Theoretical Gap Acknowledgement**: Explicitly separate the observational constraint on w from the theoretical prediction of universe fate.

Experiments

Baselines

```
```json
[
 "Planck CMB-only constraints on w (flat Λ CDM model)",
 "DES Year 3 combined probes (Clustering + Weak Lensing) constraints on w"
]
```
```

Metrics

```
```json
[
 "Statistical Uncertainty on w (σ_w)",
 "Dark Energy Figure of Merit ($\text{FoM} = 1/\sigma(w_p, w_a)$ or similar inverse covariance metric)",
 "Bias in w recovery ($|w_{\text{estimated}} - w_{\text{input}}|$ in mocks)"
]
```
```

Ablation

```
```json
[
```

"High-z Exclusion: Run analysis excluding galaxies at  $z > 1$  to test if the specific LSST capability mentioned in EV-Q050-bc54ad36646f99e2f84349ca drives the improvement.",

"Weak Lensing Only: Assess constraints using only shear data to isolate the contribution of clustering.",

"Systematics Fixed: Run with fixed photometric redshift biases to estimate the ideal statistical limit."

]

'''

## Validation Protocol

Split the mock catalog into two independent halves for cross-validation. Perform blind analysis where the input cosmology of the mock is hidden until the pipeline is finalized. Check for consistency between clustering-only and lensing-only constraints.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- \*\*EV-Q050-bc54ad36646f99e2f84349ca\*\* · arxiv · arXiv:1211.0310
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1211.0310.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED; locator=page:7|section:page-7|paragraph:1; content\_sha256=a63020139eec119e30dfcf802f79e806fd3a440fe07ff1e00cf45cd1cf85f27a

## Reviewer Comments

- Revision successfully addresses all critical issues from the previous iteration. The hypothesis no longer claims specific precision (2%) or definitive cosmological fate, correctly reframing the study as testing constraint improvement relative to baselines.
- Dataset source has been corrected to 'LSST Simulated Stacked Image Mocks' grounded in EV-Q050-bc54ad36646f99e2f84349ca, removing unverified references to CosmoDC2/OuterRim.
- Baselines now correctly cite Planck and DES Y3, removing the hallucinated 'LSST Forecast Gold sample'.
- Technical details have been sanitized of malformed LaTeX placeholders; standard cosmological notation ( $\sigma_w$ ,  $\Omega_m$ ) is now used correctly.
- The causal link between  $w$  and universe fate is explicitly treated as a knowledge gap/pending validation step rather than an established mechanism, adhering to evidence limitations.
- Ablation study comparing high- $z$  inclusion vs. exclusion has been added, directly testing the evidentiary claim regarding redshift reach.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Document the exact generation parameters for LSST simulated stacked image mocks consistent with EV-Q050-bc54ad36646f99e2f84349ca.
- Record random seeds for MCMC chains and mock generation.
- Specify prior distributions for all cosmological and nuisance parameters.
- Version control the code used for likelihood calculation and two-point statistics.
- Archive the covariance matrices used in the likelihood analysis.